

MIAE 215 · Mini-Course Lesson 5

Control Statements, Decision Logic & Loops

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering (MIAE)

Source: [Lesson 5 Web Page](#) | Video: [lesson5_control_statements.mp4](#)

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1. Decision Branching: `if` and `if-else`

Conditional structures allow programs to take alternate branches based on Boolean evaluation:

```
if (distance > 0.5) {  
    cout << "Path clear: Move robot forward  
";  
} else {  
    cout << "Obstacle detected: Stop robot  
";  
}
```

2. Relational Operators & Dynamic Conditions

- `>` (greater than), `<` (less than)
- `>=` (greater than or equal), `<=` (less than or equal)
- `==` (equal to), `!=` (not equal to)

Teacher Exam Warning / Pitfall:

Never confuse assignment (`=`) with equality (`==`):

```
if (x = 5) // BUG: assigns 5 to x; always evaluates to TRUE!  
if (x == 5) // CORRECT: tests if x is equal to 5
```

3. Boolean Logic Gates: `&&`, `||`, `!`

- `&&` (Logical AND): True only if both conditions are true.
- `||` (Logical OR): True if at least one condition is true.
- `!` (Logical NOT): Inverts condition truth value.

```
if ((temperature < 100.0) && (pressure < 250.0)) {  
    cout << "Turbine operating within normal parameters."  
};  
}
```

4. Iterative Loops: `for` and `while`

The `for` Loop

Ideal for loops with a known iteration count:

```
for (int i = 0; i < 10; i++) {  
    cout << "Iteration: " << i << "  
";  
}
```

Sentinel Termination with `break`:

```
for (int i = 0; i < 1000; i++) {  
    if (sensor_value > 500) break; // Exits loop immediately  
}
```

5. Teacher Code & Exercise Walkthroughs

- [lesson5.cpp](#): Demonstration of branching structures and loops.
- [lesson5_exercises.cpp](#) & [lesson5_exercises_solutions.cpp](#): Exercise questions and solutions.
- [lesson5_more.cpp](#) & [lesson5_more_solutions.cpp](#): Supplemental practice exercises.